

Exercise of Supervised Learning: Boosting Part 2

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July 9, 2026

Exercise 1: Gradient Boosting

In the following, you assume that your outcome follows a $\log_2$ -normal distribution with density function

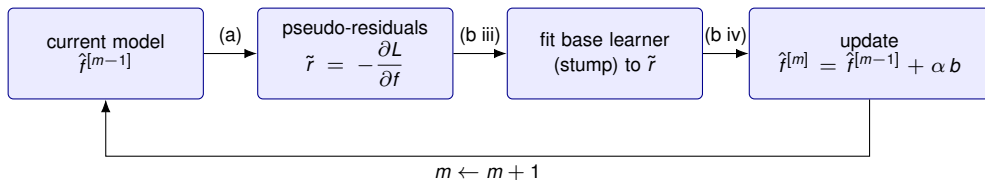
$$p(y|f) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log_2(y) - f)^2}{2\sigma^2}\right)$$

where $\sigma = 1$. In other words, $\log_2(Y)$ follows a normal distribution. You observe $n = 3$ data points $\mathbf{y}$ and want to model f using features $\mathbf{X} \in \mathbb{R}^{n \times p}$. You choose to use gradient boosting with tree base learners. (The concrete data and the $p = 2$ features are specified in part (b).)

(a) Derive the pseudo residuals based on the negative log-likelihood for the given distribution assumption.

Gradient Boosting: One Iteration at a Glance

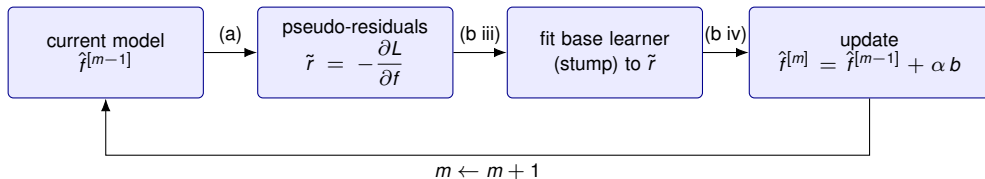
Parts (a)–(c) walk through the loop below, repeated for $m = 1, \dots, M$ rounds. At round m we fit a base learner to the *current* pseudo-residuals, then take an additive step.



(a): turn the loss into residuals (b): run one full iteration on 3 samples (c): does a second iteration help?

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1 (a): Think Before Solving

Target: turn the distribution $p(y|f)$ into a pseudo-residual $\tilde{r}$.

- ▶ **Likelihood.** The loss in gradient boosting is the NLL $L = -\ell$ with $\ell = \log p(y|f)$ — what does ℓ look like once you take the log?
- ▶ **Constants.** Which terms do not depend on f ? Why may they be dropped for the gradient?
- ▶ **Negative gradient.** Differentiate the squared term $\frac{1}{2}(\log_2(y) - f)^2$ — and recall the pseudo-residual is the *negative* gradient $\tilde{r} = -\partial L / \partial f$.

$$p(y|f) \rightarrow \ell(f) \rightarrow L = -\ell \rightarrow \tilde{r} = -\partial L / \partial f$$

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Solution to Exercise 1 (a)

(a) Derive the pseudo residuals based on the negative log-likelihood.

- Log-likelihood (drop f -independent constants):

$$\ell(f) = \log p(y|f) = \text{const} - \frac{1}{2}(\log_2(y) - f)^2.$$

- The loss is the negative log-likelihood:

$$L(y, f) = -\ell(f) = \text{const} + \frac{1}{2}(\log_2(y) - f)^2.$$

- The pseudo-residual is the *negative* gradient of the loss:

$$\tilde{r} = -\frac{\partial L(y, f)}{\partial f} = \frac{\partial \ell(f)}{\partial f} = \log_2(y) - f. \quad \square$$

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Exercise 1 (b)

(b) Given only the 3 samples $\mathbf{y} = (1, 2, 4)^\top$
and two features

$$\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2) = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}$$

explicitly derive or state with explanation

- (i) the loss-optimal initial boosting model $\hat{f}^{[0]}(\mathbf{x})$,
- (ii) the pseudo residual $\tilde{r}^{[1]}$,
- (iii) the regression stump $R_t^{[1]}$, $t = 1, 2$,
- (iv) the boosting model $\hat{f}^{[1]}(\mathbf{x})$ as well as
- (v) the pseudo residual $\tilde{r}^{[2]}$

for tree base learners with depth $d = 1$ (stumps) and a learning rate of $\alpha = 1$.

1 (b): The Data, and What One Iteration Repeats

The three samples (note $\mathbf{x}_2$ is constant $\Rightarrow$ carries no information):

i	$y^{(i)}$	$\log_2 y^{(i)}$	$x_1^{(i)}$	$x_2^{(i)}$
1	1	0	1	0
2	2	1	1	0
3	4	2	0	0

Target we regress on: $\log_2 \mathbf{y} = (0, 1, 2)^\top$.

Think before solving: what are the repeating steps of *one* gradient-boosting iteration?

- ▶ Where does the model *start* before any tree is added?
- ▶ Given the current model, how are the residuals $\tilde{r}$ formed (cf. (a))?
- ▶ A depth-1 stump can only split on *one* feature — which one, and what value sits in each leaf?
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- Initialize with the loss-minimizing constant:

$$\hat{f}^{[0]}(\mathbf{x}) = \arg \min_{f^{[0]}} \sum_{i=1}^n L(y^{(i)}, f^{[0]}(\mathbf{x}^{(i)})).$$

- Since L is the squared error on $\log_2 y$, the minimizer is the mean:

$$\hat{f}^{[0]}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \log_2(y^{(i)}) = \frac{0 + 1 + 2}{3} = 1.$$

- Hence the initial predictions on all 3 samples are

$$\tilde{\mathbf{f}}^{[0]} = (1, 1, 1)^\top. \quad \square$$

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► (i) init $\hat{f}^{[0]}$
mean of $\log_2 \mathbf{y}$

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 $\log_2 \mathbf{y} - \tilde{\mathbf{f}}^{[0]}$

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$$\tilde{f}^{[0]} = (\hat{f}^{[0]}(\mathbf{x}^{(1)}), \hat{f}^{[0]}(\mathbf{x}^{(2)}), \hat{f}^{[0]}(\mathbf{x}^{(3)}))^{\top} = (1, 1, 1)^{\top}.$$

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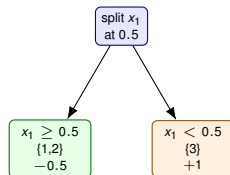
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Fit the stump to $\tilde{r}^{[1]} = (-1, 0, 1)^\top$:

- Split on x_1 (x_2 carries no info); note $x_1^{(1)} = x_1^{(2)}$.
- Each leaf value = mean residual of its samples.
- Leaf $x_1 \geq 0.5$ (samples 1,2):
 $\frac{1}{2}(-1 + 0) = -0.5$.
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$$R_1^{[1]} = -0.5 \cdot I_{x_1 \geq 0.5}, \quad R_2^{[1]} = 1 \cdot I_{x_1 < 0.5}. \quad \square$$

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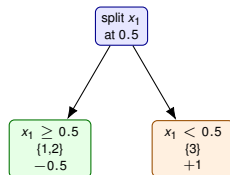
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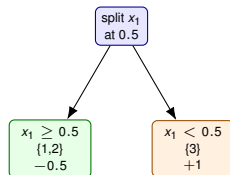
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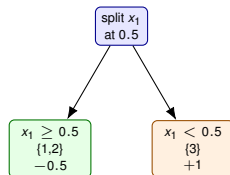
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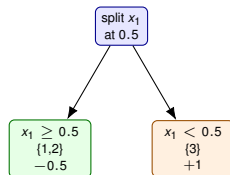
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1 (b)(iv): Boosting Model $\hat{f}^{[1]}(\mathbf{x})$ (predictions $\tilde{f}^{[1]}$)

- Recall $R_1^{[1]} = -0.5 \mathbf{I}_{x_1 \geq 0.5}$, $R_2^{[1]} = 1 \cdot \mathbf{I}_{x_1 < 0.5}$, and $\alpha = 1$.
- Update direction (per sample) from the stump: $(-0.5, -0.5, 1)^\top$.
- Additive update:

$$\begin{aligned}\tilde{f}^{[1]} &= \tilde{f}^{[0]} + \alpha \cdot (-0.5, -0.5, 1)^\top \\ &= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -0.5 \\ -0.5 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \\ 2 \end{pmatrix}. \quad \square\end{aligned}$$

Roadmap (b)

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mean of $\log_2 \mathbf{y}$

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(c) What would happen in the second iteration of the previous boosting algorithm?

Leading question: samples 1 and 2 share the same features — what value must a stump give them, and what does that do to residuals of opposite sign?

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▶ Leaf values:

$$R_1^{[2]} = \frac{-0.5+0.5}{2} I_{x_1 \geq 0.5} = 0, \quad R_2^{[2]} = 0 \cdot I_{x_1 < 0.5} = 0.$$

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these features cannot improve the model further.

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Exercise 1 (d)

(d) If you are given more data points, but still the two binary feature vectors $\mathbf{x}_1$ and $\mathbf{x}_2$, what will happen as

- (i) the number of boosting rounds M grows
- (ii) the sample size n grows

in terms of model capacity (if d is kept fixed)?

Leading question: which knob adds new base learners to the additive model — the number of rounds M , or the sample size n ?

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- (i) **M grows:** each round adds another stump (write $b^{[m]}$ for the round- m stump) to $\hat{f} = \hat{f}^{[0]} + \sum_{m=1}^M \alpha b^{[m]} \Rightarrow$ capacity *increases*, the algorithm may **overfit**.
- (ii) **n grows:** the number of base learners (depth d fixed, M fixed) is unchanged $\Rightarrow$ capacity *stays the same*, the algorithm may **underfit**.

$M \uparrow$
more stumps
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Summary — Connecting the exercise to the lecture

	What we derived	Lecture concept made concrete
(a)	Pseudo-residual $\tilde{r} = -\partial L / \partial f = \log_2(y) - f$ from the $\log_2$ -normal NLL.	Pseudo-residuals as the negative gradient of the loss.
(b)	One full iteration on 3 samples: init $\hat{f}^{[0]} = 1$, residual $\tilde{r}^{[1]} = (-1, 0, 1)^\top$, stump on x_1 , additive update $\tilde{f}^{[1]} = (0.5, 0.5, 2)^\top$, residual $\tilde{r}^{[2]} = (-0.5, 0.5, 0)^\top$.	One gradient-boosting iteration: init, residuals, fit base learner to residuals, additive update.
(c)	Second update is $(0, 0, 0)^\top$: samples 1 & 2 share features but have opposite-sign residuals $\Rightarrow$ boosting stalls.	Base learners limited by features; boosting can stall.
(d)	$M \uparrow$: capacity grows (may overfit); $n \uparrow$ (with d, M fixed): capacity unchanged (may underfit).	Capacity grows with the number of rounds M , not with n .